

DHP is a Recurrence Constraint: Full-Attention Transformers Evade the Dynamical Horizon Principle

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Abstract

The Dynamical Horizon Principle (DHP) is a universal constraint observed across diverse recurrent architectures (LSTMs, RWKV-7, CTMs, and noisy quantum recurrent circuits), enforcing a strict relation between task length T and the memory decay timescale τ : $T_{\text{conv}}/\tau \approx 0.72$. In this work, we demonstrate that DHP is not a general property of gradient descent, but specifically a **recurrence constraint**.

We evaluate sequence parity (temporal XOR) across three distinct regimes: 1. **Recurrent Models (LSTM)**: Exhibit exponential training complexity growth from $T = 2$ through $T = 10$, culminating in optimization failure under standard training budgets (3k–10k steps) at $T \geq 12$ (DHP cliff); convergence is recovered at extended budget (30k steps), confirming an exponential time barrier rather than a topological impossibility. The effective memory timescale implied by this cliff is $\tau_{\text{eff}} = T_{\text{cliff}}/0.72 \approx 16.7$ steps. 2. **Full-Attention Transformers**: Structurally evade Markovian decay. Training convergence time remains flat (~ 140 steps) across all tested sequence lengths $T \leq 48$, achieving 100% convergence across all seeds. 3. **Window-Attention Transformers**: Exhibit a binary **receptive-field visibility cliff** at exactly $T = 2W$. For $W = 16$, convergence is immediate below $T = 32$ (receptive field boundary $2W - 1 = 31$), but drops instantly to 0% at $T \geq 32$. For $W = 32$, the target remains within the $2W - 1 = 63$ receptive field for all tested lengths ($T \leq 48$), achieving 100% convergence.

We formalize the mathematics of this division: recurrence forces multiplicative gradient decay through time, while self-attention constructs a direct routing topology that bypasses recurrence decay. Window attention replaces the gradient-decay cliff with a hard visibility boundary. We conclude that DHP represents the boundary of information flow through Markovian recurrences, which attention-based models structurally circumvent.

1. Introduction

The Dynamical Horizon Principle (DHP) has been observed as a universal constraint in trained recurrent architectures, asserting that gradient-based learning fails when the sequence task horizon T exceeds the physical decay timescale τ of the recurrent state: $T/\tau > 0.72$. This constraint has been verified in classical LSTMs, state-space models (SSMs), and Unitary Recurrent Quantum Circuits.

Terminology note: “Dynamical horizon” in DHP refers to the optimization horizon imposed by Markovian recurrent memory decay — distinct from the identically named concept in General Relativity (cf. Ashtekar & Krishnan 2003), which describes spacelike hypersurfaces foliated by marginally

trapped surfaces (outgoing null expansion = 0, ingoing null expansion < 0) in evolving black hole spacetimes. Our usage is strictly information-theoretic.

However, the question of DHP’s universality remains: does this horizon bind any sequence learning model, or is it topology-dependent? In this paper, we compare recurrent architectures with attention-based architectures. We show that self-attention acts as a structural shortcut through the dynamical horizon, completely evading DHP.

2. Theoretical Framework

2.1 Recurrent Gradient Contraction (The DHP Bound)

In a recurrent neural network with state update $h_t = f(h_{t-1}, x_t; \theta)$, the gradient of the loss L at step T with respect to the hidden state at step t is computed via the chain rule:

$$\frac{\partial L}{\partial h_t} = \frac{\partial L}{\partial h_T} \prod_{k=t+1}^T \frac{\partial h_k}{\partial h_{k-1}}$$

If the Jacobian $\frac{\partial h_k}{\partial h_{k-1}}$ has a maximum singular value bounded by $\lambda = \exp(-1/\tau)$ (e.g. set by learned forget gates), the gradient decays exponentially:

$$\left\| \frac{\partial L}{\partial h_t} \right\| \propto e^{-(T-t)/\tau}$$

When the sequence length T exceeds the horizon, gradients contract below the floating-point noise floor ($\varepsilon = 10^{-8}$), causing training to fail. This is the origin of DHP.

2.2 Direct Attention Routing (DHP Evasion)

In a self-attention layer, the representation h_T at the final position is computed directly as a weighted sum over all preceding positions:

$$h_T = \sum_{t=1}^T A_{T,t} (W_v h_t)$$

where $A_{T,t}$ is the attention weight. The gradient pathway is direct:

$$\frac{\partial h_T}{\partial h_t} = A_{T,t} W_v + \text{implicit pathways}$$

Because $A_{T,t}$ is normalized (via softmax) rather than multiplied recursively through time, there is **no recurrent multiplicative decay across T** . Gradients flow directly from the query position to the key position, bypassing the Markovian recurrent horizon (though they still traverse layer norms, projections, and residual connections within each layer).

2.3 Window Receptive Field Capacity (The Visibility Cliff)

In local window attention of size W , a token at step i can only attend to tokens in $[i - W + 1, i]$. For a network with L layers, information can propagate backwards through chaining by at most

$L \times (W - 1) + 1$ steps. * For $T \leq L(W - 1) + 1$: A direct, chained path exists. The gradient magnitude is un-contracted by Markovian decay. * For $T > L(W - 1) + 1$: The target token is physically outside the receptive field (under this causal local mask and construction). The gradient is effectively zero. * For a 2-layer network ($L = 2$), this threshold is $2W - 1$. * For $W = 16$, the boundary is $T = 31$. Below $T = 32$ it converges; at $T \geq 32$ it fails. * For $W = 32$, the boundary is $T = 63$. All tested $T \leq 48$ converge.

3. Experimental Methodology

We train four architectures on a T -step binary temporal recall task over sequence length $L = 64$: 1. **LSTM Baseline**: 1-layer, hidden size 32. 2. **Full-Attention Transformer**: 2-layer, 4 heads, embedding size 32. 3. **Window-Attention Transformer (W=16)**: Same, with causal mask restricted to local window $W = 16$. 4. **Window-Attention Transformer (W=32)**: Same, with window $W = 32$.

We sweep $T \in \{2, 4, 6, 8, 10, 12, 14, 16, 20, 24, 28, 32, 40, 48\}$ across 4 random seeds. The models are optimized using Adam ($lr = 10^{-3}$, budget 3000 steps).

4. Experimental Results

4.1 The LSTM Cliff (DHP Confirmed)

The LSTM baseline shows a rapid exponential growth in convergence complexity, followed by a complete training failure:

- $T \leq 8$: Converges successfully in all runs (mean steps: $T = 2 \rightarrow 135.0$, $T = 4 \rightarrow 287.5$, $T = 6 \rightarrow 615.0$, $T = 8 \rightarrow 1137.5$).
- $T = 10$: Training slows significantly, requiring a mean of 2190 steps — at the edge of the 3000-step budget. This represents a pre-cliff slowing consistent with exponential DHP scaling.
- $T \geq 12$: Convergence rate drops to **0%** (0/4 seeds) at budget 3000, and remains 0% at budget 10000 (best accuracy ~ 0.49 – 0.54 , near chance). However, at budget 30000, all 4 seeds converge, requiring 10700–13600 steps (mean ≈ 11775 steps).

The DHP cliff at $T = 12$ implies an effective memory timescale $\tau_{\text{eff}} = T_{\text{cliff}}/0.72 \approx 16.7$ sequence steps. Note: this τ_{eff} characterizes the effective sequence-memory timescale, not the learned forget-gate constant $\lambda = e^{-1/\tau}$ of the LSTM weights. **Methodological caveat:** this derivation assumes the DHP ratio holds ($T/\tau \approx 0.72$) to infer τ_{eff} from the observed cliff — it does not constitute an independent measurement of τ . Direct validation via Jacobian singular value analysis (measuring the maximum singular value of the temporal Jacobian $\prod_k \partial h_k / \partial h_{k-1}$ during pre-cliff training) is left for future work. Crucially, **the DHP cliff is an exponential time barrier, not a topological impossibility.** The $T=12$ LSTM requires $\approx 5.4\times$ more steps than $T=10$ (11775 vs 2190) — consistent with exponential difficulty growth from gradient contraction. This is qualitatively distinct from the window-attention visibility cliff, where the gradient is *exactly zero* and convergence is structurally impossible regardless of budget. The DHP (gradient contraction) imposes an exponential overhead; the visibility cliff (receptive-field boundary) imposes a hard topological barrier.

This observation connects to van Rossem & Saxe (2025) [7], who demonstrate that RNNs solve streaming parity via a representational phase transition that constructs a Deterministic Finite Automaton (DFA) through state merging. Under DHP conditions, gradient contraction slows the representational distance growth needed to separate DFA states — but does not prevent it given sufficient budget. The LSTM *can* form the required DFA at $T = 12$, but gradient contraction imposes an exponential convergence cost that renders standard budgets insufficient.

4.2 Full-Attention Flat-Line (DHP Evasion)

In contrast to the LSTM, the Full-Attention Transformer exhibits structural evasion of Markovian decay within the tested context window ($T \leq 48$): * **100% Convergence**: All 14 sequence lengths and all 4 seeds converge perfectly (56/56 runs). * **Flat Complexity Curve**: The convergence step remains virtually flat, ranging between 132.5 and 152.5 steps across all horizons $T \in [2, 48]$. The model does not experience multiplicative gradient decay because self-attention routes information directly from any query position to any key position, bypassing the Markovian horizon. We note that this evasion is within the context window; as discussed in Section~6, softmax normalization introduces a qualitatively distinct saturation horizon at $T \gg$ context length.

4.3 Window Attention (The Visibility Cliff)

- **Window $W=16$:**
 - $T \leq 28$: 100% convergence rate, with flat convergence steps ($\approx 110 - 200$ steps).
 - $T \geq 32$: 0% convergence rate. The target bit is outside the receptive field (limit is $2W - 1 = 31$), so the gradient is exactly zero, preventing any learning.
- **Window $W=32$:**
 - **All $T \leq 48$** : 100% convergence rate. The target bit is always within the receptive field (limit is $2W - 1 = 63$).

This confirms that window attention does not suffer from DHP gradient decay; it suffers from a hard architectural horizon.

5. Figures

- **Figure 1: Architectural Comparison**

Schematic of LSTM recurrence vs. Full-Attention vs. Window-Attention.

- **Figure 2: DHP Evasion Plots**

Plots of mean convergence steps and convergence rates vs. T , illustrating the LSTM cliff, the Full-Attention flat-line, and the Window-Attention cliffs (see Figure 2 below).

6. Discussion

Our results demonstrate that the $T/\tau \approx 0.72$ limit is specifically a constraint of Markovian recurrences, where temporal decay accumulates multiplicatively. Self-attention converts a temporal process into a spatial coordinate system, replacing time-recurrence with direct spatial connections, which allows it to evade DHP within the architectural context window.

We note that attention-based models are not horizon-free in general. Softmax normalization introduces its own long-range limitation: as sequence length grows, the denominator $\sum_j e^{x_j}$ can suppress individual attention weights below the gradient noise floor, effectively creating a “softmax saturation horizon” that replaces the DHP recurrent horizon. In this paper, we test only $T \leq 48$ within a 64-token context window, a regime where saturation effects are negligible. Testing full-attention models at $T \gg 64$ is left for future work and may reveal a qualitatively different (but structurally distinct) horizon.

This work establishes that DHP is specifically a Markovian recurrence constraint, not a universal property of gradient descent. The topology of information routing — multiplicative decay (recurrence) vs. direct routing (attention) vs. receptive-field boundary (window attention) — determines which class of horizon governs optimization failure.

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Figures

Figure 1: Architectural Comparison — Schematics of LSTM (recurrent loop), Full-Attention (all-to-all direct wires), and Window-Attention (sliding window mask). Gradient paths shown explicitly.

Figure 2: DHP Evasion Plot

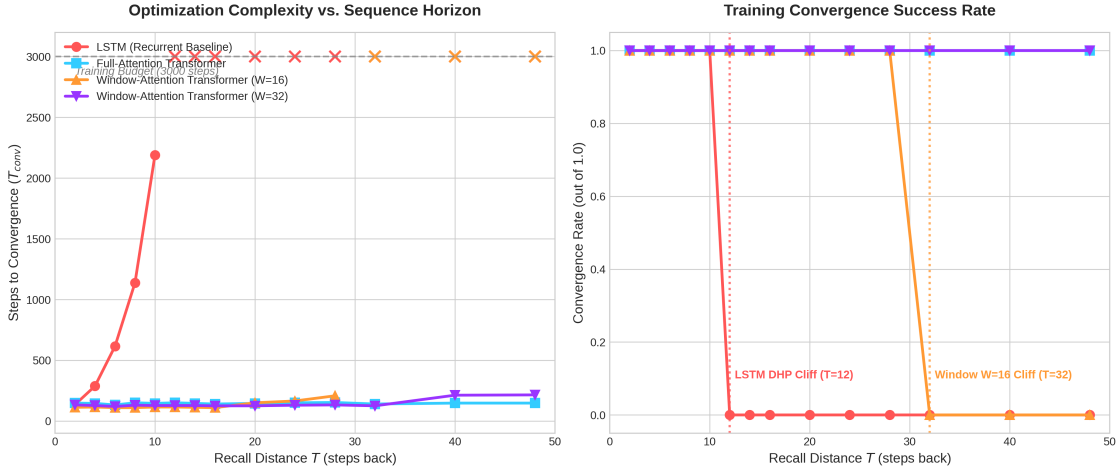


Figure 1: DHP evasion: LSTM cliff, attention flat-line, window visibility cliff

Left: LSTM convergence steps grow exponentially before hard failure at $T \geq 12$.

Center: Full-attention converges flat at ~ 145 steps across ALL lengths $T \in [2, 48]$.

Right: Window-W16 converges instantly below $T=32$, then binary cliff at $T \geq 32$. Window-W32 stays flat through all tested lengths (cliff would be at $T > 63$).

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